

Pulse-programmed mechanical memory for Sarrus-cell robotic matter

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Robotic materials need local states that can be changed briefly and then held without continuous power. FluxCell is a source-bounded design for a Sarrus-compatible cell that uses a fixed electropermanent-magnetic cartridge and moving steel keepers to make geometry, magnetic overlap, and mechanical state testable in one bench experiment. The present contribution is a locked architecture and proof protocol, not a completed actuator dataset. Force, current, heat, release, repeatability, and cycle-life claims remain unmade until the one-axis proof is built and measured.

1 Introduction

Soft robots and robotic materials increasingly distribute function through body geometry, material state, and embedded physical computation rather than through a single remote actuator (1–4). This direction is attractive because mechanical structure can store information, route signals, and shape motion before software has to intervene. The unresolved engineering question is how to make a small unit that can change state, hold that state with negligible standby power, and tile into larger visible machines without requiring a new actuator design for every morphology.

Electropermanent magnets offer a direct route to low-standby-power state retention. An electropermanent magnet combines magnetic materials with different coercivities so that a short current pulse can switch the net external field while the magnetized state persists after current returns to zero (5–7). EPs have been used for programmable matter, modular robotic connection, magnetic soft-robot components, and magnetorheological fluid control (5–8). In parallel, physical-interface work has shown that embedded magnets can create legible, passive haptic states in digitally fabricated mechanisms (9).

FluxCell connects those threads to a Sarrus-cell robotic-matter concept. The design does not ask a small magnet to pull a compliant mechanism across a large open gap. Instead, a fixed electropermanent magnet cartridge creates a localized magnetic circuit, while steel fork and tongue keepers slide through that circuit as the Sarrus-compatible frame changes width. This turns the first experiment into a compact overlap problem: pulse the cartridge, observe whether the geometry changes, check whether the state holds at zero current, reverse the pulse, and measure the limits. The paper is intentionally strict because the project is still pre-measurement. Its novelty is the architecture and falsifiable proof plan; performance claims are deferred until data exist.

2 Results

2.1 A fixed cartridge turns actuation into keeper overlap

The locked architecture separates the magnetic cartridge from the moving carriers (Fig. 1). The cartridge contains the Alnico pulse element, the NdFeB bias element, and the top and bottom steel pole plates. The moving structure contributes steel keepers only: two fork prongs from one side and a center tongue from the opposite side. The useful magnetic path is therefore local to the overlap between the fixed pole plates and the moving steel keepers. This makes the one-axis proof inspectable because the visible width change and the hidden magnetic variable are linked by one measurable dimension: keeper overlap.

Where the EPM is: fixed center cartridge; prongs/tongues are moving steel keepers

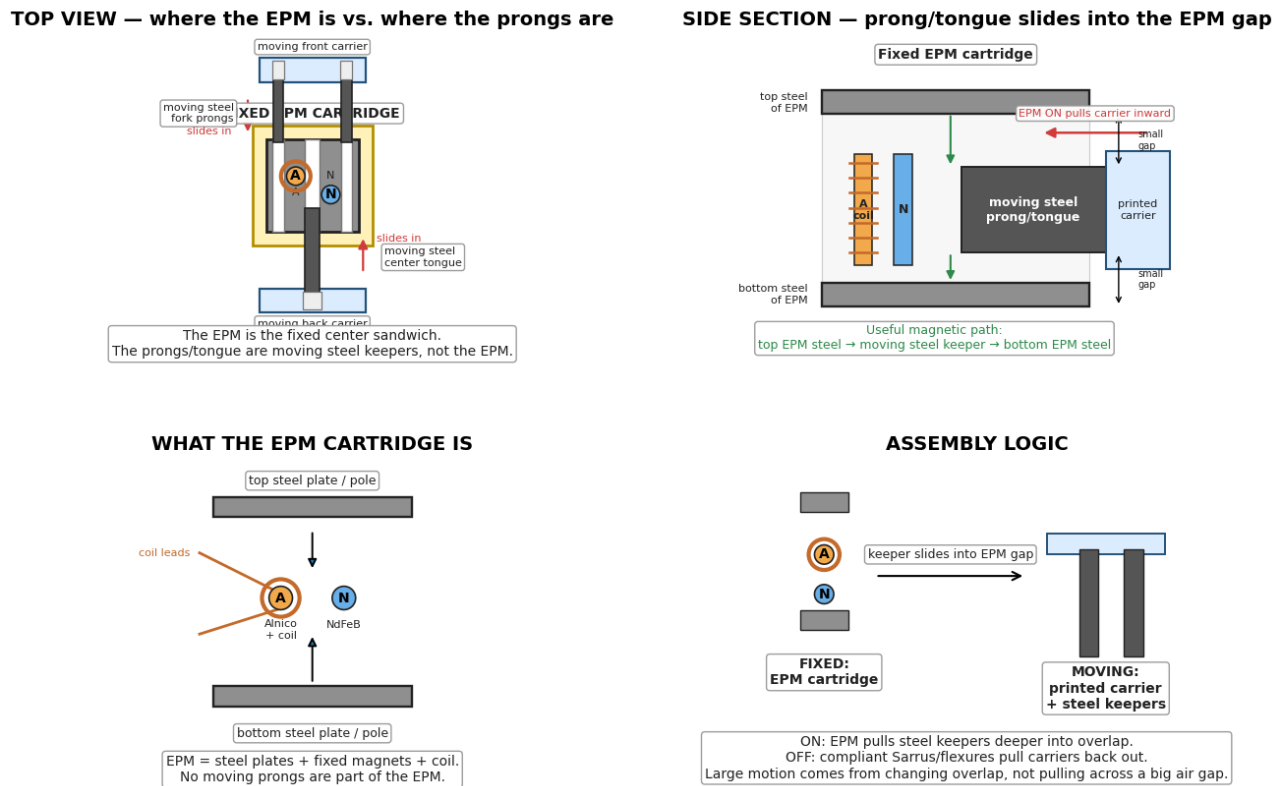


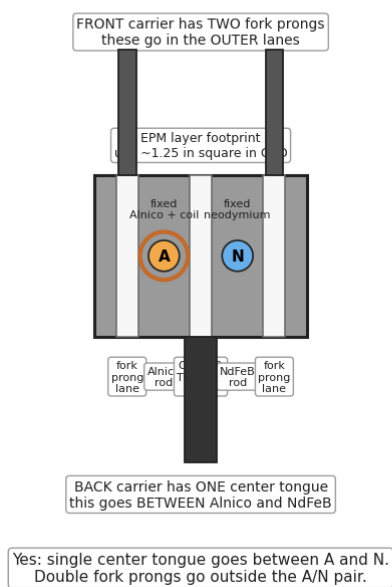
Figure 1. Fixed cartridge and moving keeper concept. The EPM cartridge remains fixed while fork prongs and a center tongue move through the working gap. The useful magnetic path is top pole plate to moving steel keeper to bottom pole plate. Source: FluxCell uploaded design figure, 13 May 2026.

2.2 The lane layout removes collision before fabrication polish

The one-axis layer uses a comb/fork topology (Fig. 2). The center tongue enters between the Alnico and NdFeB rods, while the fork prongs pass outside the pair. The moving steel pieces therefore interdigitate rather than collide. This geometry is a prerequisite for meaningful actuation testing: if the carriers collide, any later force or current measurement would only describe a packaging failure. The lane figure therefore serves as an engineering gate before committing to a monolithic two-axis cell.

Clear lane layout: center tongue between Alnico/NdFeB, fork prongs outside

ONE AXIS LAYER — LANE ORDER



ASSEMBLED TOP VIEW — WIDE/OFF vs PULLED-IN

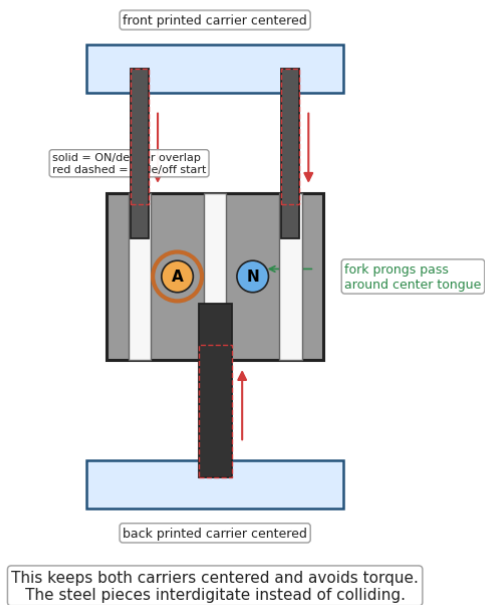
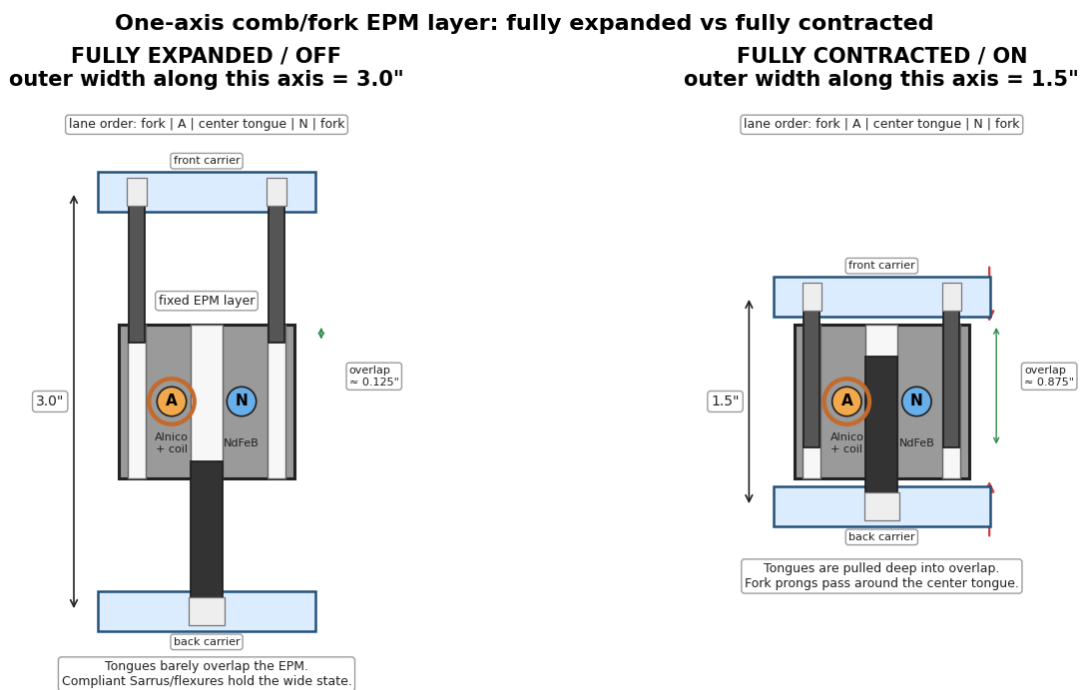


Figure 2. Comb/fork lane order. The center tongue runs between the Alnico and NdFeB positions, and fork prongs run on the outside lanes. The layout keeps both carriers centered and gives the magnetic circuit a repeatable overlap variable. Source: FluxCell uploaded design figure, 13 May 2026.

2.3 The dimensional lock creates a falsifiable binary proof

The current dimensional correction uses 1.00 inch keeper inserts rather than 1-1/8 inch inserts. With 1.25 inch square pole plates, the one-axis proof targets a 3.0 inch expanded/off width and a 1.5 inch contracted/on width, corresponding to a 0.75 inch side stroke. The off-state overlap is approximately 0.125 inch and the on-state overlap is approximately 0.875 inch (Fig. 3). These numbers are design targets, not measured performance. Their value is that they make the first physical test falsifiable: a synchronized record can show width, pulse polarity, current, hold at zero drive, release after reverse pulse, and coil temperature.



This shows one axis only. The second axis is the same layer rotated 90° and stacked at a different height.

Figure 3. One-axis overlap target. The expanded/off and contracted/on states create a large visible width change while the magnetic variable stays local to keeper overlap. Current evidence locks the geometry; measured force, current, heat, repeatability, and release margins remain pending until the proof is built. Source: FluxCell uploaded design figure, 13 May 2026.

2.4 The build lock is small enough to purchase and test

Table 1. Locked purchase logic for one proof plus four complete two-axis cells. Costs are purchase-planning values before shipping and tax, not experimental performance data.

Part	Qty	Locked specification	Approx total
Alnico 5 rods	12	0.250 inch diameter by 0.625 inch long, axial	\$23.88
NdFeB rods	12	K&J D4A, 1/4 inch diameter by 5/8 inch long, N42 axial	~\$15.00
Square pole plates	16	1018 steel, 0.25 inch thick by 1.25 inch wide, cut length 1.25 inch	~\$56.00
Fork prong inserts	24	A36 flat bar, 1/8 inch by 1/2 inch, cut length 1.00 inch	\$167.28
Center tongue inserts	12	1018 flat bar, 1/4 inch by 1/2 inch, cut length 1.00 inch	\$87.72
Wire, tape, supply, leads, pigtails, heat shrink	set	30 AWG magnet wire, Kapton, 0–30 V / 0–10 A supply, leads, silicone wire, strain relief	~\$106.75

The total before shipping and tax is approximately \$456.65 (10–13). This number is not a claim about manufacturability at scale. It is a friction claim: the proof has a specific bill of materials and no longer depends on an undefined part-search step.

3 Discussion

The strongest version of FluxCell is not a generic magnetic actuator paper. It is a paper about pulse-programmed mechanical memory. The state variable is not merely magnetic polarity; it is a visible mechanical configuration coupled to magnetic overlap and held after the electrical pulse ends. If the one-axis proof records clean switching, zero-current hold, reverse-pulse release, bounded heating, and repeated cycles, the work can move from design protocol to actuator evidence.

The source boundary matters. This manuscript does not claim force density, efficiency, safety, lifetime, or reliable two-axis behavior. It claims that the current design has become concrete enough to test and that the proof has been reduced to a small set of measurements. That reduction is the paper's present contribution. It makes the next experiment difficult to hand-wave: either the geometry moves and holds under a measured pulse, or the data identify the dominant failure mode.

For R&D in physical experiences, the compelling artifact would be a small surface that changes

contour, holds while electrically quiet, and returns on command. That future demonstrator would be human-readable because the motion, latch, and reset are visible. The current paper does not claim that demonstrator exists. It defines the cell-level experiment that must pass before that story is credible.

4 Materials and Methods

4.1 One-axis proof geometry

The proof uses a fixed center EPM cartridge with two 1.25 inch square pole plates made from 0.25 inch by 1.25 inch 1018 steel rectangle bar cut to 1.25 inch length. The magnetic elements are one 0.250 inch diameter by 0.625 inch long Alnico 5 rod and one 1/4 inch diameter by 5/8 inch long NdFeB rod. The moving keepers are two A36 fork prong inserts, each 1/8 inch by 1/2 inch by 1.00 inch, and one 1018 center tongue insert, 1/4 inch by 1/2 inch by 1.00 inch.

4.2 Electrical pulse protocol

The Alnico coil is insulated with Kapton, wound with 30 AWG magnet wire, and connected to an H-bridge driver from a current-limited bench supply. The first test uses short bidirectional pulses separated by zero-drive intervals. For each pulse, the record should include polarity, pulse duration, current limit, observed width, hold state after zero drive, release state after reverse pulse, and a coil temperature note.

4.3 Acceptance boundary

The first proof succeeds only if the one-axis assembly visibly changes width, holds a state after current returns to zero, releases or reverses under the opposite pulse, and stays within a temperature range safe enough for continued bench testing. A partial result remains useful if it identifies the dominant failure mode: weak magnetic force, high friction, carrier twist, thermal limit, or poor release margin.

5 Acknowledgements

This draft was assembled from the FluxCell design log, uploaded schematic figures, and the current locked purchase list.

6 Funding

No external funding is claimed in this draft.

7 Author contributions

Alan N. Pham conceived the project direction, maintained the design log, selected the current purchase lock, and defined the proof target. Codex assisted with manuscript formatting, page integration, figure placement, and verification workflow.

8 Competing interests

The author declares no competing interests for this draft.

9 Data availability

The current data are design records, uploaded schematic figures, supplier links, and purchase specifications. Measured actuation data are pending the one-axis proof.

10 Code availability

The current project page, generated proof bundle, and manuscript source are served through the FluxCell site. Arduino pulse code is included in the generated proof bundle.

11 Additional information

This is a draft manuscript and design protocol, not a completed experimental report. The claims should be updated only after the one-axis proof records measured current, width, temperature, hold, release, and repeatability.

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